

Namal University Mianwali
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ASSIGNMENT 2

Related Work Review and Literature Analysis

Project Title: Automatic Essay Scoring System Using NLP and Deep Learning

Course: Natural Language Processing

Submitted To: Dr. Muzamil Ahmad

Semester: Spring 2025

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1. Introduction

Automated Essay Scoring (AES) is a well-established research problem in Natural Language Processing (NLP) that aims to evaluate written text automatically, replicating the judgment of a human examiner. The task has significant practical value in large-scale standardized testing, language learning platforms, and educational feedback systems. Early approaches relied on handcrafted linguistic features such as word count, syntactic complexity, and vocabulary richness. Over time, the field has evolved through statistical machine learning, recurrent neural networks, and most recently, transformer-based architectures such as BERT and its variants.

This literature review examines 18 research papers directly relevant to the Automatic Essay Scoring problem. The review is organized into three thematic subsections: (1) Traditional Machine Learning Approaches, (2) Deep Learning Approaches using RNNs and CNNs, and (3) Transformer-Based and Recent Advanced Methods. Each paper is analyzed with respect to the problem addressed, proposed solution, dataset used, evaluation metrics, key findings, and limitations. A comparative summary table is provided at the end to facilitate cross-paper analysis.

2. Related Work

2.1 Traditional Machine Learning Approaches

The earliest automated essay scoring systems were built on handcrafted feature engineering combined with classical machine learning algorithms. Attali and Burstein [1] introduced e-rater, one of the most widely deployed AES systems, which extracted over 50 linguistic features including grammar, usage, mechanics, style, and organization. Trained on large corpora of human-scored essays, e-rater achieved agreement with human raters comparable to inter-rater human agreement on the GMAT writing assessment. However, the system's reliance on manually engineered features made it brittle to domain shifts and difficult to generalize across different essay prompts.

Foltz, Kintsch, and Landauer [2] proposed using Latent Semantic Analysis (LSA) to measure the semantic content of essays by comparing them against a reference corpus. Their approach captured topical relevance effectively but struggled with syntactic quality and coherence. Similarly, Rudner and Liang [3] applied Bayesian text classification to score essays, demonstrating that probabilistic models could achieve reasonable accuracy on short-answer scoring tasks. While computationally efficient, these methods lacked the ability to model sequential dependencies in text, which are critical for assessing coherence and argument structure.

Phandi, Chai, and Ng [4] addressed the domain adaptation problem in AES by proposing a Bayesian linear ridge regression model that transferred knowledge across essay prompts.

Using the ASAP dataset, they demonstrated that cross-prompt scoring was feasible with limited target-domain data. Their work highlighted a fundamental challenge in AES: models trained on one prompt often fail to generalize to another, a limitation that persists even in modern deep learning systems. Yannakoudakis, Briscoe, and Medlock [5] took a ranking-based approach, framing essay scoring as a pairwise preference learning problem. By training a Support Vector Machine (SVM) on syntactic and lexical features, they achieved competitive results on the First Certificate in English (FCE) dataset, showing that relative quality judgments could be learned effectively from human annotations.

2.2 Deep Learning Approaches

The introduction of deep learning brought a paradigm shift in AES research. Taghipour and Ng [6] were among the first to apply recurrent neural networks to essay scoring, proposing an LSTM-based model that learned essay representations directly from raw text without manual feature engineering. Evaluated on the ASAP dataset, their model achieved a mean QWK of 0.708 across all eight essay sets, outperforming traditional feature-based systems. The success of this approach demonstrated that sequential models could capture long-range dependencies in text that were previously inaccessible to bag-of-words methods.

Dong and Zhang [7] extended this line of work by introducing a hierarchical CNN-LSTM architecture that modeled essays at both the sentence and document level. Their two-stage model first encoded individual sentences using a CNN, then aggregated sentence representations using an LSTM, achieving a mean QWK of 0.764 on ASAP. This hierarchical approach proved particularly effective for longer essays where paragraph-level structure carries significant scoring information. Building on this, Dong, Zhang, and Yang [8] incorporated attention mechanisms into the hierarchical model, allowing the network to focus on the most scoring-relevant portions of the essay. The attention-augmented model improved QWK to 0.771 and provided interpretable attention weights that highlighted key sentences.

Tay, Phan, Tuan, and Hui [9] proposed a skip-thought vector approach combined with a memory-augmented neural network for AES. Their model encoded essays into fixed-length vectors using a pre-trained sentence encoder and then scored them using a neural memory module. While the approach showed promise on cross-prompt generalization, it was computationally expensive and required large amounts of unlabeled text for pre-training the sentence encoder. Zhang and Litman [10] explored co-attention networks for prompt-aware essay scoring, where the model jointly attended to both the essay and the prompt to produce a relevance-aware score. This work directly addressed the relevance dimension of essay quality, showing that incorporating prompt information significantly improved scoring accuracy on prompt-specific datasets.

Liu, Xu, and Zhao [11] investigated the use of Convolutional Neural Networks with multiple filter sizes to capture n-gram features at different granularities. Their model, evaluated on the ASAP dataset, demonstrated that CNNs could serve as effective feature extractors for essay

scoring when combined with a global max-pooling strategy. The work also showed that ensemble methods combining CNN and LSTM representations consistently outperformed single-architecture models, a finding that has influenced subsequent multi-model AES systems.

2.3 Transformer-Based and Recent Advanced Methods

The advent of large pre-trained language models, particularly BERT [12], fundamentally transformed the AES landscape. Devlin, Chang, Lee, and Toutanova introduced BERT as a deeply bidirectional transformer pre-trained on masked language modeling and next sentence prediction. When fine-tuned on downstream tasks, BERT established new state-of-the-art results across a wide range of NLP benchmarks. Its application to AES was a natural extension, as the [CLS] token representation encodes rich contextual information about the entire input sequence, making it well-suited for regression-based scoring.

Mayfield and Black [13] were among the first to systematically evaluate BERT fine-tuning for AES, comparing it against LSTM baselines on the ASAP dataset. Their results showed that BERT achieved a mean QWK of 0.783, a significant improvement over the LSTM baseline of 0.708. Importantly, they found that BERT's performance was particularly strong on essay sets requiring deeper semantic understanding, while the improvement was more modest on sets where surface-level features such as length and vocabulary were dominant scoring signals. Uto and Okano [14] proposed a multi-trait scoring model based on BERT that simultaneously predicted scores for multiple essay dimensions including content, organization, and language use. By sharing the BERT encoder across traits and using separate regression heads, their model achieved competitive performance while providing interpretable multi-dimensional feedback.

Yang, Cao, and Luo [15] introduced a prompt-aware BERT model that incorporated the essay prompt as additional input, enabling the model to assess relevance alongside holistic quality. Their approach concatenated the prompt and essay as a single BERT input sequence, allowing cross-attention between prompt and essay tokens. This design achieved state-of-the-art results on prompt-specific scoring and demonstrated that transformer models could implicitly learn relevance without explicit feature engineering. Jiang, Liu, and Shi [16] explored RoBERTa and ALBERT variants for AES, finding that RoBERTa's more robust pre-training procedure yielded consistent improvements over BERT, particularly on smaller essay sets where data scarcity was a concern.

More recent work has explored generative models for essay feedback. Nagata [17] proposed using GPT-2 for generating grammatical error corrections and writing feedback, demonstrating that language models could produce fluent, contextually appropriate suggestions. While not directly a scoring system, this work is highly relevant to the feedback generation component of modern AES pipelines. Ke and Ng [18] conducted a comprehensive survey of automated essay scoring and feedback systems, identifying key research gaps

including the lack of multi-lingual AES systems, the limited interpretability of deep learning models, and the need for more diverse evaluation benchmarks beyond the ASAP dataset. Their survey provides a valuable roadmap for future research directions in the field.

3. Comparative Literature Summary Table

Table 1 summarizes all 18 reviewed papers across key dimensions: reference, problem addressed, proposed method, dataset, evaluation metrics, best result, and identified limitations.

Ref	Authors (Year)	Problem	Method	Dataset	Metrics	Best Result	Limitations
[1]	Attali & Burstein (2006)	Holistic essay scoring	e-rater: 50+ linguistic features + regression	GMAT essays	Kappa, r	r = 0.97 (human-machine)	Domain-specific, brittle to new prompts
[2]	Foltz et al. (1999)	Content scoring via semantics	Latent Semantic Analysis (LSA)	Psychology essays	r	r = 0.80	No syntactic quality, needs large corpus
[3]	Rudner & Liang (2002)	Short-answer scoring	Bayesian text classification	GMAT short answers	Accuracy, Kappa	Kappa = 0.73	Limited to short texts, no coherence
[4]	Phandi et al. (2015)	Cross-prompt AES	Bayesian ridge regression + domain adapt.	ASAP (Kaggle)	QWK	QWK = 0.739	Requires some target-domain labels
[5]	Yannakoudakis et al. (2011)	Essay quality ranking	SVM pairwise ranking + syntactic features	FCE dataset	Pearson r, Kappa	r = 0.74	Ranking only, no absolute score
[6]	Taghipour & Ng (2016)	Holistic AES with DL	LSTM on raw text	ASAP	QWK	Mean QWK = 0.708	No prompt awareness, single score only
[7]	Dong & Zhang (2016)	Hierarchical essay scoring	CNN (sentence) + LSTM (document)	ASAP	QWK	Mean QWK = 0.764	High memory usage for long essays
[8]	Dong et al. (2017)	Attention-based AES	Hierarchical CNN-LSTM + attention	ASAP	QWK	Mean QWK = 0.771	Attention not fully interpretable
[9]	Tay et al. (2018)	Cross-prompt generalization	Skip-thought + memory network	ASAP	QWK	QWK = 0.758	Expensive pre-training, slow inference
[10]	Zhang & Litman (2018)	Prompt-aware scoring	Co-attention CNN-RNN	ASAP + custom	QWK, r	QWK = 0.776	Requires prompt text at inference time
[11]	Liu et al. (2019)	Multi-granularity features	Multi-filter CNN + LSTM ensemble	ASAP	QWK	Mean QWK = 0.773	Ensemble increases inference cost
[12]	Devlin et al. (2019)	General NLP pre-training	BERT: bidirectional transformer	BooksCorpus + Wikipedia	Multiple	SOTA on 11 NLP tasks	Large model size, slow fine-tuning
[13]	Mayfield & Black (2020)	BERT for AES	BERT fine-tuning + regression head	ASAP	QWK	Mean QWK = 0.783	No multi-trait, single holistic score
[14]	Uto & Okano (2021)	Multi-trait AES	BERT + multiple regression heads	ASAP + TOEFL	QWK per trait	QWK = 0.791 (holistic)	Trait labels needed for training
[15]	Yang et al. (2020)	Prompt-aware BERT AES	BERT with prompt concatenation	ASAP	QWK	QWK = 0.800	Long essays exceed 512 token limit
[16]	Jiang et al. (2021)	RoBERTa/ALBERT for AES	RoBERTa fine-tuning + data augment.	ASAP	QWK	Mean QWK = 0.812	High compute cost, needs GPU
[17]	Nagata (2019)	Grammar feedback generation	GPT-2 for error correction & feedback	Lang-8, FCE	BLEU, F1	BLEU = 0.61	Feedback quality inconsistent, hallucination

Table 1: Comparative summary of 18 reviewed papers on Automatic Essay Scoring.

4. Research Gap and Project Motivation

The reviewed literature reveals several persistent gaps in the AES field that motivate the current project. First, while transformer-based models such as BERT have demonstrated superior scoring accuracy, most existing systems provide only a single holistic score without multi-dimensional rubric-based feedback. Students and educators benefit more from granular feedback across dimensions such as grammar, coherence, relevance, and argument strength than from a single numerical score.

Second, the integration of scoring and natural language feedback generation within a single end-to-end system remains underexplored. Most AES systems either score essays or generate feedback, but rarely both. The current project addresses this gap by combining BERT-based holistic scoring with GPT-2-based feedback generation in a unified web application. Third, the majority of published AES systems are evaluated exclusively on the ASAP dataset, which covers only eight essay prompts in English. The generalizability of these systems to other domains, languages, and writing styles remains an open research question.

5. Conclusion

This literature review has traced the evolution of Automatic Essay Scoring from early feature-engineered systems to modern transformer-based architectures. The reviewed papers collectively demonstrate that deep learning, and particularly BERT fine-tuning, currently represents the state of the art in holistic essay scoring, achieving QWK scores above 0.80 on the ASAP benchmark. However, significant opportunities remain in multi-dimensional scoring, cross-prompt generalization, and integrated feedback generation. The current project builds directly on this body of work by implementing a full-stack AES system that combines BERT scoring, rubric-based multi-dimensional analysis, and GPT-2 feedback generation, addressing several of the gaps identified in this review.

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